A High-Precision GNSS SAR Imaging Fusion Method Utilizing Optimally Matched Satellites Calculated by CRLB

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***Abstract*— The Global Navigation Satellite System (GNSS) offers advantages such as all-weather operability and extensive spatial coverage. Utilizing GNSS-reflected signals for ground synthetic aperture radar (SAR) imaging presents a cost-effective and widely applicable technical solution. However, the small bandwidth of GNSS signals results in inadequate resolution, posing challenges for practical applications. To address this issue, an SAR fusion imaging system model is established, consisting of multiple satellites and a single ground-fixed GNSS receiver. The relationship between the ambiguity function of GNSS signals and Fisher information is investigated, allowing for the derivation of the Cramer–Rao lower bound (CRLB) for the system, which is primarily influenced by the geometrical configuration of the bistatic setup. Subsequently, the CRLB expression is employed to identify the optimal resolution direction of the satellites for ground targets, and a dual-satellite SAR imaging fusion method based on optimal matching is proposed. The effectiveness of this approach is validated through simulations and real experimental data, demonstrating that the theoretically optimal resolution direction predicted by the CRLB aligns with the actual imaging results. Furthermore, the proposed method achieves higher resolution compared to traditional techniques, with the fused imaging results demonstrating a clear correspondence with the satellite imagery of the scene map.**

***Index Terms*— Cramer–Rao lower bound (CRLB), Fisher information, Global Navigation Satellite System (GNSS), image fusion, synthetic aperture radar (SAR) imaging.**

I. Introduction

GLOBAL Navigation Satellite System (GNSS) is a complex technology that enables global positioning, navigation, and time synchronization through satellite constellations. An innovative application of GNSS-reflected signal is synthetic aperture radar (SAR) imaging technology, known as GNSS-R SAR. The system typically uses a ground-fixed GNSS dual-channel receiver to capture both the transmitted signal from the satellite and the reflected signal from the target. Since no dedicated radar transmitter is required, the system is low-cost, all-weather, has wide coverage, and is currently used in a range of fields including sea surface altimetry, wind speed estimation, target detection, and surface imaging [1], [2], [3], [4], [5].

Generally, in the GNSS-R SAR system, range compression is achieved through the robust autocorrelation characteristics of the pseudo random noise (PRN) code, while azimuth compression is accomplished by leveraging satellite orbit parameters and receiver coordinates. By employing a GNSS software receiver to demodulate direct wave signals, one can extract carrier phase information, instantaneous Doppler shifts, and code delays, which subsequently facilitate the generation of reference signals for range compression [6]. In recent years, SAR imaging using GNSS-R technology has been applied to a variety of scenarios [7], [8], [9]. However, due to the limited bandwidth of GNSS signals, practical applications encounter challenges associated with low resolution. For example, the bandwidth of a GPS L1 C/A code signal is only 2.046 MHz, which results in a theoretical range resolution of approximately 150 m. This limitation significantly restricts its applicability in SAR imaging.

To address the issue of low resolution in GNSS-R SAR systems, existing solutions can be classified into three main categories: 1) selection of a high-bandwidth signal source; 2) extension of observation duration; and 3) enhancement of algorithms. The first category involves selecting GNSS signals with larger bandwidths. For instance, Zeng et al. [10] utilize the L5 signal from GPS satellites as the transmission signal. The L5 signal, with a bandwidth of approximately 20 MHz, results in a range resolution of about 15 m, enabling clear differentiation of main buildings in SAR imaging of a university campus. Similarly, Ma et al. [11] and Zeng et al. [12] employ Galileo and BeiDou satellites as transmission signals, respectively, achieving varying levels of imaging effectiveness. The second category focuses on acquiring

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extended observation data to enhance azimuth resolution. According to SAR imaging theory, longer synthetic apertures in the azimuth direction yield higher corresponding azimuth resolution. For example, Yang et al. [13] propose increasing the observation duration to improve detection accuracy in landslide detection scenarios. The third category involves improving existing algorithms to enhance system resolution. For instance, Fang et al. [14] employ an improved second-order differential algorithm to enhance the range resolution of SAR imaging. Santi et al. [15] utilize multisatellite image fusion technology to achieve higher resolution for ground scenes.

While these solutions improve system resolution in specific scenarios, each comes with inherent limitations. Based on the regulations of the International Telecommunication Union and the technical implementation considerations of GNSS systems, the existing GNSS C/A code signals have relatively narrow bandwidths [16], resulting in lower range resolution for GNSS-SAR systems. Additionally, collecting excessively long observation data escalates demands on hardware storage capabilities and increases computational load, posing challenges to cost-effectiveness and rapid detection. In terms of algorithms, noncoherent fusion of imaging results from multiple satellites offers a flexible method to enhance system resolution, leveraging high resolution in the azimuth direction for improved fidelity across the entire detection area. And detection characteristics of GNSS-R bistatic SAR systems concerning ground targets depend on the system's geometrical configuration. However, when the relative geometrical configuration between the selected satellites and the receiver is similar, their detection performance for the target also remains similar, significantly diminishing the effectiveness of fusion processing.

Existing research has shown that using noncoherent fusion processing on SAR images obtained from multiple satellites can significantly improve image resolution [17], [18], [19]; however, the issue of introducing artifacts with this method also intensifies as the number of satellites increases [20]. Since images obtained from satellites located at different positions exhibit varying orientations in their responses, the resolution of SAR images can be effectively improved by selecting satellites from appropriate positions for incoherent fusion, while minimizing the influence of artifacts as much as possible.

In general, methods based on incoherent image fusion inevitably involve the issue of selecting imaging satellites, and there is little existing research that thoroughly explores this problem. Based on the model of the GNSS-R SAR system, studying resolution performance from the perspective of the ambiguity function represents a quantifiable research method that aids algorithm improvement and parameter optimization. Wu et al. [21] investigate the ambiguity function of bistatic SAR and extend its application to the spatial domain, analyzing the system's range and azimuth resolution capabilities. Daout et al. [22] introduce the concept of the minimum resolution ellipse for the first time and provide a mathematical expression for the ambiguity function. To address evident coupling issues between azimuth and range directions in geostationary SAR (GEOSAR), Chen et al. [23] propose a resolution evaluation method based on local features of the ambiguity function. These studies primarily stem from mathematical models and can be utilized to assess system resolution under specific conditions, thereby identifying potential performance bottlenecks. However, an analysis solely from the perspective of the ambiguity function does not yield a quantitative relationship between the bistatic geometrical configuration and resolution accuracy. In the GNSS-R bistatic SAR system, the -3-dB resolution curve corresponding to ground targets represents a projection of the spatial resolution ellipsoid onto the ground, typically approximated as an ellipse. Investigating the relationship between this resolution ellipse and the bistatic geometrical configuration provides a theoretical basis for satellite selection in SAR imaging.

To address the issue of insufficient satellite selection criteria in multisatellite incoherent SAR image fusion technology, the Cramer–Rao lower bound (CRLB) for estimating ground target parameters is derived based on the geometrical configuration of the bistatic SAR imaging system. Additionally, a high-precision imaging approach for dual-satellite image fusion is proposed in this article, utilizing an optimal matching technique. Using GPS L1 C/A code signals as the transmission source, a model for the GPS SAR imaging system is established. Based on the derived CRLB expression, a 1-D search over the angle is conducted to determine the optimal resolution direction. Among all tracked GPS satellites, two satellites are selected to ensure that the angle between them at the ground's optimal resolution direction is as close to 90° as possible. Subsequently, SAR image processing and fusion are performed for the two satellites to achieve the final imaging results. The proposed method enhances imaging outcomes effectively without necessitating redundant computations, and both the simulation and experimental results substantiate its efficacy. The contributions of this work are as follows.

1) We are the first to derive the CRLB for ground target parameter estimation in a GNSS bistatic SAR imaging system. We define the ground targets within a polar coordinate framework and obtain the decoupled ambiguity function for time delay and Doppler using Taylor expansion theory. By utilizing the relationship between the ambiguity function and Fisher information, we analyze the influence of satellite orbital parameters and bistatic geometrical configuration on the CRLB. Once the satellite orbital parameters are determined, the system's resolution performance limits for the same target in different ground directions can be theoretically calculated.
2) We propose a dual-satellite SAR imaging fusion method based on optimal matching. Utilizing the derived CRLB expression, we perform a 1-D search over the angles to determine the deflection angle corresponding to the minimum CRLB value. This process identifies the optimal resolution direction for all satellites within the scene. Subsequently, based on the principle that the angle between the optimal resolution directions should be as close to 90° as possible, we select the two

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optimally matched satellites and fuse the imaging results obtained from them.

3) We are the first to construct a GPS-TAG to simulate the reflected signals of point targets in a real environment and to validate the derived CRLB using real data from the GPS-TAG. The GPS-TAG consists of a directional transmitting antenna, a GPS omnidirectional receiving antenna, and a GPS signal amplifier. The experimental results show that the theoretically optimal resolution direction calculated based on the CRLB aligns with the actual imaging results of the GPS-TAG, and the proposed method achieves the minimum −3-dB resolution area.
4) We use a left-handed circularly polarized antenna to receive GPS L1 C/A code-reflected signals from a real scene. Through SAR imaging of the real scene, we validate the effectiveness of the proposed dual-satellite imaging fusion method based on optimal matching, and we compare the fused imaging results with satellite photographs of the scene. The experimental results indicate that the proposed method significantly enhances imaging resolution performance, and the fused SAR images achieve clear alignment with the scene photographs.

The remaining sections of this article are arranged as follows. Section II introduces the proposed model and imaging principles, Section III derives the CRLB expression, Section IV presents the proposed new fusion SAR imaging method, Section V validates the proposed scheme using simulation and real experimental data, and Section VI concludes this article.

II. GPS SAR Imaging System

In this section, the geometrical configuration of the GPS SAR imaging system is defined and the GNSS signal model is established. Following this, the process of range compression and the back-projection (BP) algorithms for signal processing in SAR imaging are introduced.

A. System Model

In the 3-D Cartesian coordinate system, the SAR imaging system comprises the GPS satellite T_r, a dual-antenna receiver Rx, and a stationary ground target P, as depicted in Fig. 1. The raw measurement data are broken down into a 2-D array consisting of fast time and slow time [10], assuming that the position of the satellite is fixed during the fast time. $\widehat{OT_r}$ is the unit vector from the origin to the satellite T_r, which is given by

$$\widehat{OT_r}(\eta) = \frac{\overrightarrow{T_r}}{\|T_r\|} = \left(x^{T_r}(\eta), y^{T_r}(\eta), z^{T_r}(\eta)\right) \tag{1}$$

where η is the slow time, $\overrightarrow{T_r}$ represents the vector from the coordinate origin to the receiver, and $\|\cdot\|$ represents the modulo operation. $x^{T_r}(\eta)$, $y^{T_r}(\eta)$, and $z^{T_r}(\eta)$ are the coordinates of the satellite on the x-, y-, and z-axes, respectively. $\widehat{OR_x}$ is the unit vector from the origin to the receiver Rx, and the direction of the vector $\overrightarrow{\beta}$ is the angle between $\widehat{OT_r}$ and $\widehat{OR_x}$, which are defined as follows:

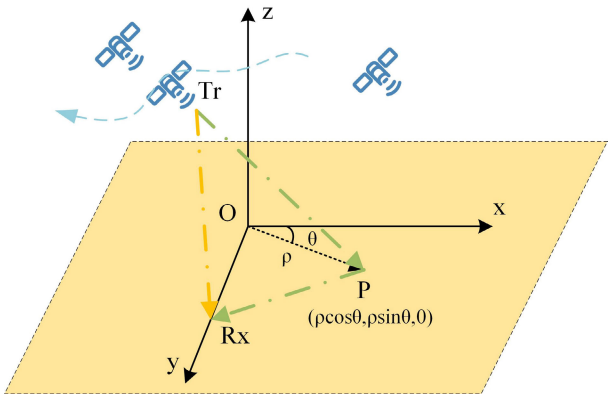


Fig. 1. GPS SAR imaging system defined in this article: satellite, ground-fixed receiver, and ground target.

$$\widehat{OR_x} = \frac{\overrightarrow{R_x}}{\|R_x\|} = (0, 1, 0) \tag{2}$$

$$\overrightarrow{\beta}(\eta) = \widehat{OT_r}(\eta) + \widehat{OR_x} \tag{3}$$

where $\overrightarrow{R_x}$ represents the vector from the coordinate origin to the receiver.

B. Signal Model and Imaging Algorithm

The GPS L1 signals can be characterized by the carrier data code and the PRN code [24]. It is assumed that the GPS L1 C/A code signals in the system are represented as follows:

$$s(t) = C(t) \times D(t) \times \cos(2\pi f_c t + \varphi_0) \tag{4}$$

where the PRN code (denoted by C) is unique for each satellite, the data code (denoted by D) conveys the navigation message of the respective satellite, and f_c and φ_0 are the carrier frequency and initial phase, respectively. To adapt to the SAR imaging model, the original 1-D GNSS data are organized into 2-D based on fast time and slow time. Given that the period of the C/A code signals is 1 ms, the interval for slow time is also set to 1 ms. The following formulas represent 2-D direct waves and 2-D reflected waves, respectively:

$$s_d(\eta, \tau) = C\left[\tau - \frac{D_d(\eta)}{c}\right] \exp\left(-j2\pi \frac{D_d(\eta)}{\lambda}\right) \tag{5}$$

$$s_r(\eta, \tau) = \sigma C\left[\tau - \frac{D_r(\eta)}{c}\right] \exp\left(-j2\pi \frac{D_r(\eta)}{\lambda}\right) \tag{6}$$

where τ is the fast time, σ is the scattering coefficient of target, λ is wavelength, and c is the light speed. D_d and D_r represent the distances traveled by the direct wave and the reflected wave, respectively. In a slow time period, the distance is considered to be a constant value.

In light of the nonlinear trajectory of GPS satellites, the BP imaging algorithm effectively mitigates errors associated with approximation operations. Its robustness renders it impervious to variations in the radar's motion trajectory, making it suitable for a wide array of imaging scenarios [25], [26].

Due to the strong autocorrelation characteristics of the PRN code, a pure reference signal can be reconstructed from

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the demodulation information of the direct wave for the purpose of cross correlation with reflected signals [27]. In this context, $s_d(\eta,\tau)$ denotes the direct wave, $s_r(\eta,\tau)$ represents the reflected wave, and $s_{rc}(\eta,\tau)$ signifies the data after range compression, which can be expressed as

$$s_{rc}(\eta,\tau)=F^{-1}\left\{F\left[s_d^*(\eta,\tau)\right]\cdot F[s_r(\eta,\tau)]\right\} \tag{7}$$

where $F[\cdot]$ and $F^{-1}[\cdot]$ specifically refer to the Fourier transform and inverse Fourier transform in the fast-time direction. For a certain point (x_i, y_i) in the scene, $(x_i, y_j)\in\Omega$, Ω is a set of scene coordinates. Let its scattering coefficient be $f(x_i, y_i)$. According to the principles of the BP algorithm, $f(x_i, y_i)$ can be obtained by integrating the range compressed data in the azimuth direction, and the final imaging result can be expressed as [28], [29]

$$f(x_i,y_j)=\int_0^T s_{rc}\left(\eta_m,\tau_{ij}(\eta_m)\right)\exp\left(j2\pi f_c\frac{\Delta R_{ij}(\eta_m)}{c}\right)d\eta_m \tag{8}$$

where T is the synthetic aperture time, and η_m represents a certain slow-time moment. $\tau_{ij}(\eta_m)=(\Delta R_{ij}(\eta_m)/c)$ represents the time delay, and $\Delta R_{ij}(\eta_m)$ represents the differential distance between the direct wave and the reflected wave. For the scattering coefficient $f(x, y)$ of the entire scene, simply repeat the operations of (8) for each coordinate point. In practical applications, the coordinates of the satellite at each slow-time moment can be derived from the demodulated navigation message. The grid point coordinates and satellite coordinates of the imaging scene are subsequently converted to East–North–Up (ENU) coordinates, using the receiver as the origin.

III. CRLB Derivation

CRLB serves as a theoretical limit for assessing the accuracy of estimators [30]. In the context of the GNSS-R SAR imaging system, the resolution characteristics fundamentally represent a method for estimating target parameters. The resolution performance of the system has a direct impact on the outcomes of target localization and imaging. The ambiguity function is defined as the cross-correlation function of the echoes from two adjacent targets. Assuming points O and P are on the ground, with O at the origin, the point target is defined using polar coordinates to more effectively represent the accuracy of target position estimation across various angular directions. The polar coordinates of point P are expressed as $(\rho\cos\theta, \rho\sin\theta, 0)$, and (9) represents the time-domain representation of the ambiguity function corresponding to points O and P. By utilizing Parseval's theorem and the properties of the Fourier transform [22], the frequency-domain representation of the ambiguity function can be obtained, as shown in (10)

$$\chi(\overrightarrow{OP})=K\int\int s_O(\eta,\tau)s_p^*(\eta,\tau)d\eta d\tau \tag{9}$$

$$\chi(\overrightarrow{OP})=K\int\int|\widetilde{P}(f-f_c)|^2\exp\left(-j\frac{2\pi f}{c}\overrightarrow{OP}\cdot\overrightarrow{\beta}(\eta)\right)\times\exp\left(j\frac{2\pi f_c}{c}\overrightarrow{OP}\cdot\overrightarrow{\beta}(\eta)\right)df d\eta \tag{10}$$

where s_O represents the echo at point O, s_p represents the echo at point P, and $|\widetilde{P}(\cdot)|^2$ is the power spectrum of the signal. $\eta\in[0, T]$. K is the normalization coefficient. According to the deduction in the Appendix, it can be seen that the change of K exerts the same influence on each angle direction and does not affect the results of the subsequent 1-D search for the angle. For the sake of clarity in the following paragraphs, the influence caused by the change of K is omitted. In (10), f and η are coupled seriously, which brings inconvenience to the subsequent operation.

For GPS L1 C/A code signals, the bandwidth B_{L1} is 2.046 MHz, so f and η can be decoupled only by the system parameters that satisfy $c/(2T\overrightarrow{OP}\cdot(d\overrightarrow{\beta}/d\eta)(\eta_0))\geq B_{L1}$, and the decoupled ambiguity function can be expressed as (see the Appendix)

$$\chi(\overrightarrow{OP})=K\times B\frac{\sin(\pi B\tau_d(\rho,\theta))}{\pi B\tau_d(\rho,\theta)}\times T\frac{\sin(\pi T f_d(\rho,\theta))}{\pi T f_d(\rho,\theta)}. \tag{11}$$

In (11), $\tau_d(\rho,\theta)$ and $f_d(\rho,\theta)$ represent the time delay and Doppler frequency generated by the target P in the system, respectively.

According to the relationship between ambiguity function and Fisher information [31], the Fisher information matrix $\mathbf{J}_{\tau_\mathbf{d},\mathbf{f_d}}$ with respect to τ_d and f_d can be given by

$$\mathbf{J}_{\tau_\mathbf{d},\mathbf{f_d}}=\mathrm{SNR}\begin{bmatrix}J_{11}&J_{12}\\J_{21}&J_{22}\end{bmatrix} \tag{12}$$

where SNR represents the signal-to-noise ratio. Since $\tau_d(\rho,\theta)$ and $f_d(\rho,\theta)$ are related to vector $\overrightarrow{OP}$, $\chi(\overrightarrow{OP})$ is replaced with $\chi(\tau_d(\rho,\theta), f_d(\rho,\theta))$ in the following discussion. And the Fisher information matrix $\mathbf{J}_{\tau_\mathbf{d},\mathbf{f_d}}$ can be expressed in terms of the performance of the ambiguity function at the origin, denoting as

$$J_{11}=-\left.\frac{\partial^2\chi(\tau_d(\rho,\theta),f_d(\rho,\theta))}{\partial\tau_d^2}\right|_{\tau_d=0,f_d=0}=\frac{B^3KPT\pi^2}{3} \tag{13}$$

$$J_{22}=-\left.\frac{\partial^2\chi(\tau_d(\rho,\theta),f_d(\rho,\theta))}{\partial f_d^2}\right|_{\tau_d=0,f_d=0}=\frac{BKPT^3\pi^2}{3}. \tag{14}$$

In (11), the Doppler frequency is decoupled from the time delay, and it follows that:

$$J_{12}=J_{21}=0. \tag{15}$$

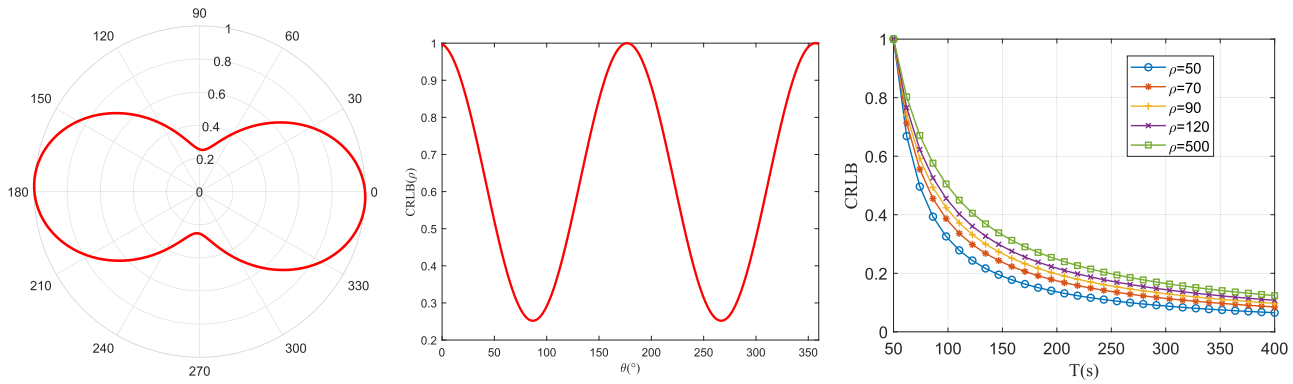
Substituting the coordinates of point P into (A.6) and (A.7), the following results are obtained:

$$\tau_d(\rho,\theta)=\frac{\rho\cos\theta x^{T_r}(\eta_0)+\rho\sin\theta\left(y^{T_r}(\eta_0)+1\right)}{c} \tag{16}$$

$$f_d(\rho,\theta)=\frac{v_x\rho\cos\theta+v_y\rho\sin\theta}{\lambda_0} \tag{17}$$

where f_0 is the carrier frequency and λ_0 is the wavelength of the carrier. The instantaneous velocity of the satellite at η_0 is

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(a) (b) (c)

Fig. 2. Changes in CRLB under different conditions. (a) Variation of CRLB values with different direction angles. (b) Results of a 1-D search for angles. (c) Effects of varying ρ and T on CRLB.

defined as

$$v_x = \left.\frac{d(x^{T_r}(\eta))}{d\eta}\right|_{\eta=\eta_0} \tag{18}$$

$$v_y = \left.\frac{d(y^{T_r}(\eta))}{d\eta}\right|_{\eta=\eta_0}. \tag{19}$$

And the definitions for $\overline{\eta} = [\tau_d, f_d]$ and $\widehat{\eta} = [\rho, \theta]$ are established, with the Jacobian matrix $\mathbf{\Psi}$ of $\overline{\eta}$ with respect to $\widehat{\eta}$ expressed as $\Psi = \partial\overline{\eta}^T/\partial\widehat{\eta}$. Based on the properties of Fisher information, as referenced in [32], [33], and [34], the Fisher information matrix for the parameter $\widehat{\eta}$ is obtained and denoted as $\mathbf{J}_{\rho,\theta}$, given by

$$\mathbf{J}_{\rho,\theta} = \mathbf{\Psi}\mathbf{J}_{\tau_d,f_d}\mathbf{\Psi}^{\mathbf{T}}. \tag{20}$$

To streamline the final outcome, the following variables are defined:

$$t_1 \triangleq \frac{\partial\tau_d(\rho,\theta)}{\partial\rho} = \frac{x^{T_r}(\eta_0)\cos\theta + \left(y^{T_r}(\eta_0)+1\right)\sin\theta}{c} \tag{21}$$

$$t_2 \triangleq \frac{\partial\tau_d(\rho,\theta)}{\partial\theta} = \frac{-x^{T_r}(\eta_0)\rho\sin\theta + \left(y^{T_r}(\eta_0)+1\right)\rho\cos\theta}{c} \tag{22}$$

$$f_1 \triangleq \frac{\partial f_d(\rho,\theta)}{\partial\rho} = \frac{v_x\cos\theta + v_y\sin\theta}{\lambda_0} \tag{23}$$

$$f_2 \triangleq \frac{\partial f_d(\rho,\theta)}{\partial\theta} = \frac{-v_x\rho\sin\theta + v_y\rho\cos\theta}{\lambda_0} \tag{24}$$

$$\Gamma \triangleq {t_1}^2 J_{11} + {t_2}^2 J_{22} + t_1 t_2 J_{12} + t_1 t_2 J_{21} \tag{25}$$

$$\mathrm{E} \triangleq t_1 f_1 J_{11} + t_2 f_1 J_{21} + t_1 f_2 J_{12} + t_2 f_2 J_{22} \tag{26}$$

$$\mathrm{H} \triangleq f_1^2 J_{11} + f_1 f_2 J_{21} + f_1 f_2 J_{12} + f_2^2 J_{22}. \tag{27}$$

Based on (21)–(27) and the relationship between Fisher information matrix and CRLB, the CRLB expression for the parameter ρ is obtained after matrix inversion and is expressed as follows:

$$\mathrm{CRLB}(\rho) = \frac{\mathrm{SNR}^{-1}}{\Gamma - \mathrm{E}^2/\mathrm{H}}. \tag{28}$$

Based on (28), the following conclusions can be drawn.

1) CRLB(ρ) is related to SNR, bandwidth (B), synthetic aperture time (T), the target's polar angle (θ), satellite position, and velocity.
2) Under the same condition, the value of CRLB(ρ) is different for different angle directions.

For the convenience of subsequent analysis, CRLB is used instead of CRLB(ρ). To analyze the influence of the relevant variables in (28) on CRLB, numerical analysis is performed according to the parameters of G1 satellite and receiver in Table I.

Fig. 2(a) shows the changes of CRLB with different direction angles in the form of polar coordinates. It can be seen from the figure that there are two directions that make CRLB have the maximum value and the minimum value, respectively, and these two directions are vertical. This corresponds to the major axis and the minor axis of the point target minimum resolution ellipse in the bistatic SAR system. Fig. 2(b) gives the results of CRLB 1-D search for directional angle, and the results of this 1-D search are consistent with Fig. 2(a). In practical application, the optimal resolution direction of the target on the ground can be determined directly according to the results of the angle search. Fig. 2(c) shows how the minimum value of CRLB changes with ρ and T. For the purpose of comparison, the CRLB presented in the figure has been normalized. It is evident from Fig. 2(c) that the CRLB increases as the values of ρ and T decrease. Notably, when T is larger, the CRLB value is smaller, which aligns with the principles of actual SAR imaging, specifically, a longer radar synthetic aperture time results in higher azimuth resolution.

IV. Proposed Method

Building upon the research presented in Section III, the optimal ground resolution direction for each satellite can be identified by calculating the CRLB. From the collected satellite data, two satellites are selected whose optimal resolution direction differences are closest to 90° for SAR imaging and incoherent fusion. This strategy maximizes computational efficiency while enhancing imaging effectiveness. The CRLB expression and the optimal resolution angle of the ith satellite are defined as $\mathrm{CRLB}(\rho)^i$ and θ^i, respectively, satisfying $\theta^i = \arg\min_\theta\{\mathrm{CRLB}(\rho)^i\}$. The two satellites ith and jth calculated according to the following equation represent the optimally matched satellites selected,

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denoted as:

$$i^*, j^* = \arg\min_{i,j}\left\{\left|\left|\theta^i - \theta^j\right| - 90\right|\right\}. \tag{29}$$

The proposed imaging scheme is illustrated in Fig. 3, which primarily consists of two stages: satellite selection and imaging fusion. A dual-channel GNSS receiver is used to simultaneously acquire direct wave and reflected wave signals, with the waveform of the raw data depicted in the figure. In the satellite selection stage, GPS satellites are first captured and tracked using the GNSS software receiver, and the orbital parameters of each satellite are calculated based on the direct wave signal. Subsequently, the ENU coordinates are converted from the receiver's latitude and longitude coordinates and the imaging grid points, allowing for the selection of the two optimally matched satellites according to the principles outlined in (29). In the imaging fusion stage, the basic parameters of the reference signal, including the PRN code, carrier phase, Doppler shift, code delay, and navigation message, are first generated using the tracking results of the direct wave signal. Then, cross-correlation processing is performed between the reference signal and the reflected wave signal to achieve range compression. Finally, the imaging area is divided into a grid, and the scattering coefficient for each grid point is computed using (8). This process results in the SAR imaging outcomes for the entire scene. The SAR images corresponding to satellite i^* and satellite j^* are represented as $f_1(x, y)$ and $f_2(x, y)$, respectively. The fusion of these SAR images entails normalizing $f_1(x, y)$ and $f_2(x, y)$ before summing them, as illustrated in the following equation:

$$f(x, y) = \frac{f_1(x, y) - \min\{f_1(x, y)\}}{\max\{f_1(x, y)\} - \min\{f_1(x, y)\}} + \frac{f_2(x, y) - \min\{f_2(x, y)\}}{\max\{f_2(x, y)\} - \min\{f_2(x, y)\}} \tag{30}$$

where $\max\{\cdot\}$ and $\min\{\cdot\}$ represent the maximum and minimum values, respectively. Finally, the SAR image fusion results $f(x, y)$ of two optimally matched satellites can be obtained, and a simulation of the point target fusion process is also presented in the figure.

V. Simulation and Experimental Analysis

In this section, simulations and real experimental data are employed to validate the accuracy of the derived CRLB and to evaluate the effectiveness of the dual-satellite optimal matching approach in SAR imaging fusion.

A. Simulation and Analysis

To validate the effectiveness of the proposed method, four discrete point targets on the ground, designated as P1, P2, P3, and P4, are defined based on the system model illustrated in Fig. 1. Two satellites, G1 and G2, are established by following the principles outlined in (28). According to the theoretical values calculated by the CRLB, the optimal resolution angles for satellites G1 and G2 are determined to be 90° and 176°, respectively. The angle between the optimal resolution directions on the ground is approximately 86°. Additional relevant simulation parameters are detailed in Table I. The entire simulation experiment is conducted using MATLAB 2021b.

TABLE I
Simulation Parameters

Parameter	Value
G1 Position	$[9.0 \times 10^6, 1.4 \times 10^7, 1.2 \times 10^7]m$
G1 Velocity	$[-1822, 103, 2243]m/s$
G2 Position	$[8.2 \times 10^6, 1.9 \times 10^7, 7.1 \times 10^6]m$
G2 Velocity	$[-47, -1103, 2024]m/s$
Rx Position	$[0, 500, 0]m$
P1 Position	$[40, 40, 0]m$
P2 Position	$[-40, 40, 0]m$
P3 Position	$[-40, -40, 0]m$
P4 Position	$[40, -40, 0]m$
PRF	$1000Hz$
Sampling Frequency	$30MHz$
Synthetic aperture duration	$60s$
Satellite Signal	GPS L1 C/A code

The SAR imaging results of the two satellites are presented in Fig. 4(a) and (b). As illustrated in the figure, the limited bandwidth of the GPS L1 C/A code signals results in poor range-oriented resolution, whereas the longer synthetic aperture time enhances azimuth-oriented resolution. Consequently, the four point targets in Fig. 4(a) and (b) cannot be fully distinguished. The SAR imaging results of the two points along the range direction exhibit aliasing. Analysis of the −3-dB imaging results reveals that the optimal resolution direction of the system aligns with the theoretically calculated value. The fusion process of the images shown in Fig. 4(a) and (b) is depicted in Fig. 4(c), while the −3-dB imaging results following image fusion are presented in Fig. 4(d). And Fig. 4(e) shows the 3-D stereoscopic results after image fusion. The fused image clearly delineates the four point targets, with their central positions closely approximating the theoretical positions.

The simulation results indicate that the proposed method significantly enhances the resolution performance of the system, effectively compensating for the limitations associated with the bandwidth of the GPS L1 signals.

B. GPS-TAG Imaging Experiment and Analysis

To validate the derived expression of the CRLB in a real experimental environment, a GPS-TAG was designed to simulate GPS-reflected signals from a point target. The GPS-TAG is defined as a type of radio frequency tag that consists of an omnidirectional GPS receiving antenna, a GPS signal low-noise amplifier, and a directional transmitting antenna, primarily functioning as a signal relay. Upon receiving the satellite's direct wave signal, the GPS-TAG locally amplifies the original signal with low noise before transmitting it through the transmitting antenna. This operational approach effectively simulates the scenario of a target reflecting the GPS signals in a real environment. Given that the intensity of the GPS-TAG's transmitted signal is significantly greater than that of the GPS-reflected signal in the surrounding environment and the satellite's direct wave signal, the influence of noise can be considered negligible. Theoretically, GPS-TAG SAR imaging results in a single isolated point.

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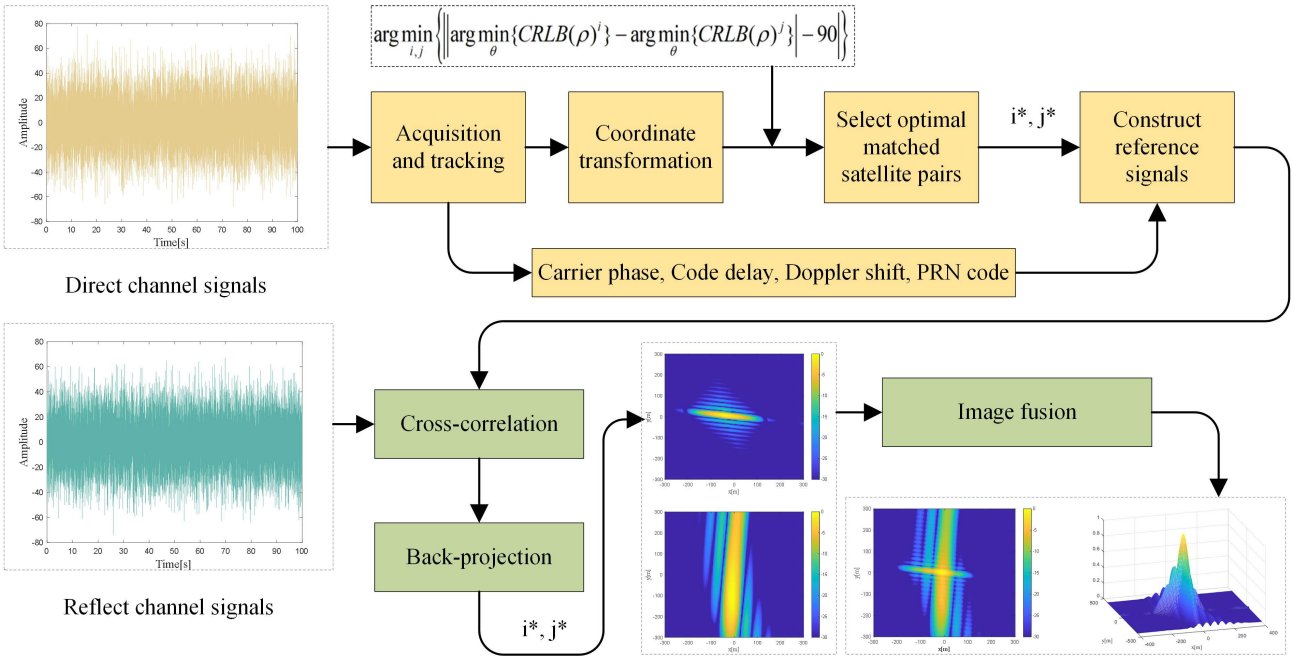
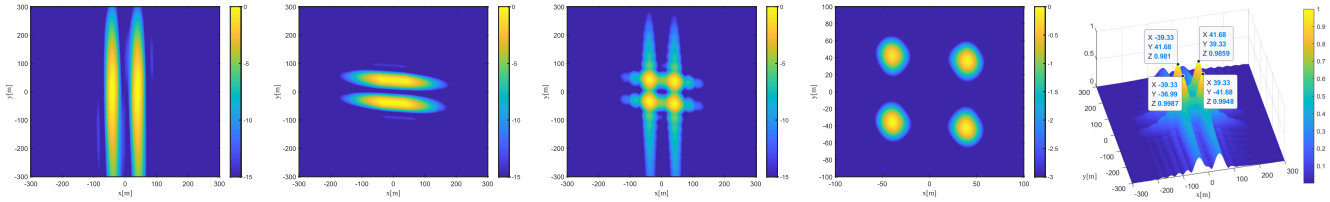


Fig. 3. Proposed flowchart for optimal matching in dual-satellite fusion imaging.



(a) (b) (c) (d) (e)

Fig. 4. Result of simulation experiment. (a) BP SAR imaging of satellite G1. (b) BP SAR imaging of satellite G2. (c) Image fusion (−15 dB). (d) Image fusion (−3 dB). (e) 3-D stereogram from fused images.



Fig. 5. Experimental setup of the GPS-TAG and the experimental scene.

The experimental setup and equipment are illustrated in Fig. 5, and the experiments are conducted at Chongqing University of Posts and Telecommunications, Chongqing, China. The receiver is positioned on the roof of the Yifu Building, while the GPS-TAG is located on the roof of No. 4 teaching building, to the East of the Yifu Building. The control of the USRP N321 is implemented through GNU Radio. The processed raw sample data are handled by a GNSS software receiver to capture and track GPS satellites. Two experiments are conducted independently for GPS L1 C/A code and L5 C/A code signals, with sampling rates set at 15.36 and 100 MHz, respectively, and data collection durations of 2 and 4 min. Ultimately, after processing by the GNSS software receiver, the GPS-TAG experiment for the GPS L1 C/A code signals successfully tracks four GPS satellites: G15, G18, G24, and G32. Similarly, the GPS-TAG experiment for the GPS L5 C/A code signals tracks four GPS satellites: G17, G23, G01, and G03. The latitude and longitude of the receiver, along with the satellite coordinates and imaging grid coordinates, are converted into the ENU coordinate system, facilitating SAR imaging processing based on the conditions of each experiment.

Fig. 6 presents the BP SAR imaging results of the GPS-TAG obtained from all satellites, alongside the theoretical CRLB values as a function of the deflection angle. Fig. 6(a)–(d) presents the experimental results for the GPS L1 C/A code signals, while Fig. 6(e)–(h) shows the experimental results for the GPS L5 C/A code signals. The image in the lower right corner of each figure serves as a reference for comparison with the corresponding results. The deflection angle associated with the minimum CRLB value indicates the optimal resolution direction of the bistatic system formed by the selected satellites on the ground, while the deflection angle corresponding to the maximum CRLB value reflects the worst resolution achievable by the system. Evidently, for both the GPS L1 C/A code and GPS L5 C/A code signals, the

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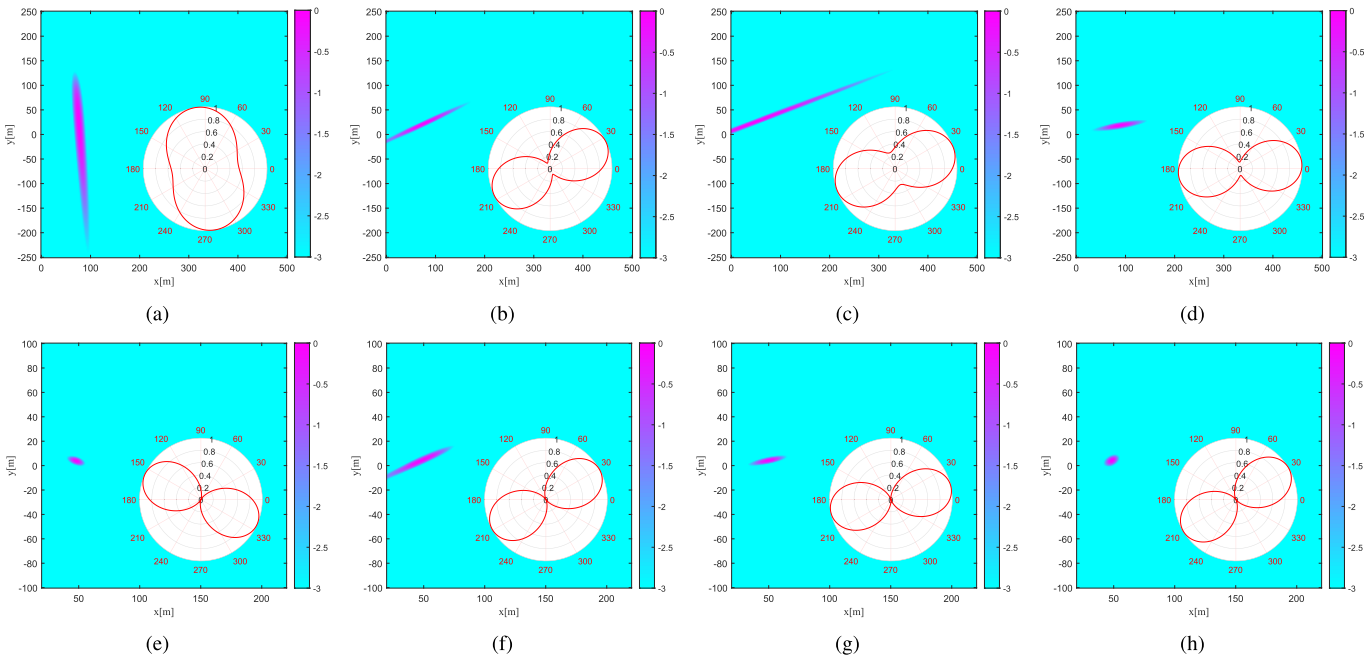


Fig. 6. BP SAR imaging results (−3 dB) from various satellites in the GPS-TAG experiment, accompanied by a polar coordinate diagram illustrating the corresponding CRLB as the angle varies. (a) G15. (b) G18. (c) G24. (d) G32. (e) G17. (f) G23. (g) G01. (h) G03.

optimal resolution direction observed in the −3-dB imaging results for each satellite aligns closely with the theoretically calculated direction. This consistency validates the accuracy of the CRLB in predicting the optimal performance of SAR imaging.

To further investigate the imaging fusion results of various satellite pairs, fusion processing is performed on the imaging results from all combinations of satellites. Fig. 7 presents the fusion results for all satellite pair images, with the bottom-right image in each figure displaying the corresponding −3-dB results. Fig. 7(a)–(f) presents the experimental results for the GPS L1 C/A code signals, while Fig. 7(g)–(l) shows the experimental results for the GPS L5 C/A code signals. In the experimental results for the GPS L1 C/A code signals, the optimal resolution angles of satellites G15 and G32 are closest to 90°, resulting in the best fusion outcome depicted in Fig. 7(c), characterized by the smallest resolution area. In contrast, the optimal resolution directions of satellites G18 and G24 are similar, leading to comparable results for the G15–G18 and G15–G24 combinations. The optimal resolution directions of the three satellites G18, G24, and G32 are relatively close, which adversely affects the fusion quality of all their combinations, as illustrated in Fig. 7(d)–(f). In the experimental results for the GPS L5 C/A code signals, the optimal resolution angles of satellites G17 and G03 are closest to 90°, resulting in the best fusion outcome depicted in Fig. 7(i), characterized by the smallest resolution area. Similarly, the results for other satellite combinations also deteriorate as the proximity to their respective optimal resolution direction increases.

Table II presents the experimental results for each pair of satellites, detailing the theoretical angle difference of the optimal resolution direction, the actual angle difference, and the minimum resolution unit area. Given that the theoretical imaging outcome of GPS-TAG is represented as an isolated point, we define the minimum circular area of −3-dB region fit as the area corresponding to the minimum resolution unit. It is evident that the actual angle differences for each group of satellite pairs closely align with the theoretically calculated values. Furthermore, a trend is observed: as the angle difference approaches 90°, the area of the minimum resolution unit decreases, resulting in enhanced resolution in the fused imaging results. Since the bandwidth of the GPS L5 C/A code signals is greater than that of the GPS L1 C/A code signals, the overall resolution of the GPS L5 signals is superior to that of the GPS L1 signals.

Figs. 8 and 9 illustrate the results of SAR image fusion using combinations of three and four satellites, respectively. Figs. 8(a)–(d) and 9(a) present the experimental results for the GPS L1 C/A code signals, while Figs. 8(e)–(h) and 9(b) show the experimental results for the GPS L5 C/A code signals. It is evident that compared to the fusion results obtained from two satellites, the integration of multiple satellite data does not yield a significant improvement in resolution. In the experiment of GPS L1 C/A code signals, false target is appeared, and this finding parallels the simulation outcomes reported in the literature [20], which also noted the emergence of ghost targets during the incoherent fusion process involving multiple satellites. Particularly, as evidenced in Fig. 8(a), (c), and (d), the quality of the fusion result diminishes. Although there is no ghosting observed in Fig. 8(b), its minimum resolution unit area is still smaller than that of the result shown in Fig. 7(c). In the experiments with the GPS L5 C/A code signals, the broader bandwidth and longer code length of the GPS L5 C/A code signals, as well as its better correlation performance, lead to improved suppression of false targets caused by sidelobe overlap during the fusion process. As a result, there are almost no false

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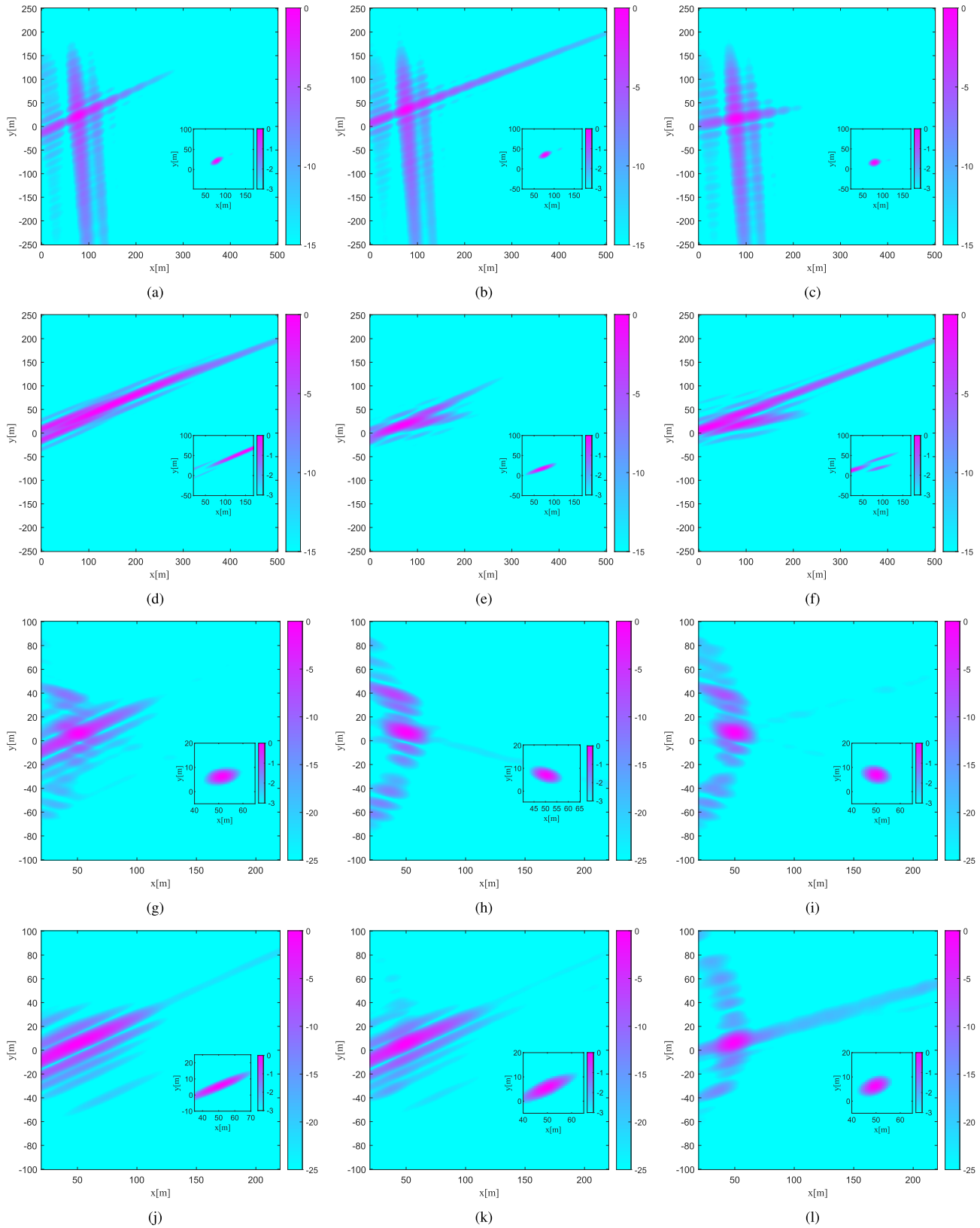


Fig. 7. Image fusion of two satellite combinations. (a) G15–G18. (b) G15–G24. (c) G15–G32. (d) G18–G24. (e) G18–G32. (f) G24–G32. (g) G17–G23. (h) G17–G01. (i) G17–G03. (j) G23–G01. (k) G23–G03. (l) G01–G03.

targets observed in Figs. 8(e)–(h) and 9(b). However, due to factors such as coordinate errors of different satellites and receivers, as well as phase errors [35], the center positions of the main lobes in the SAR imaging results may vary slightly. This variation leads to a certain degree of broadening and defocusing of the main lobes during the incoherent fusion of multiple SAR images, which can also result in a degradation of the fusion quality.

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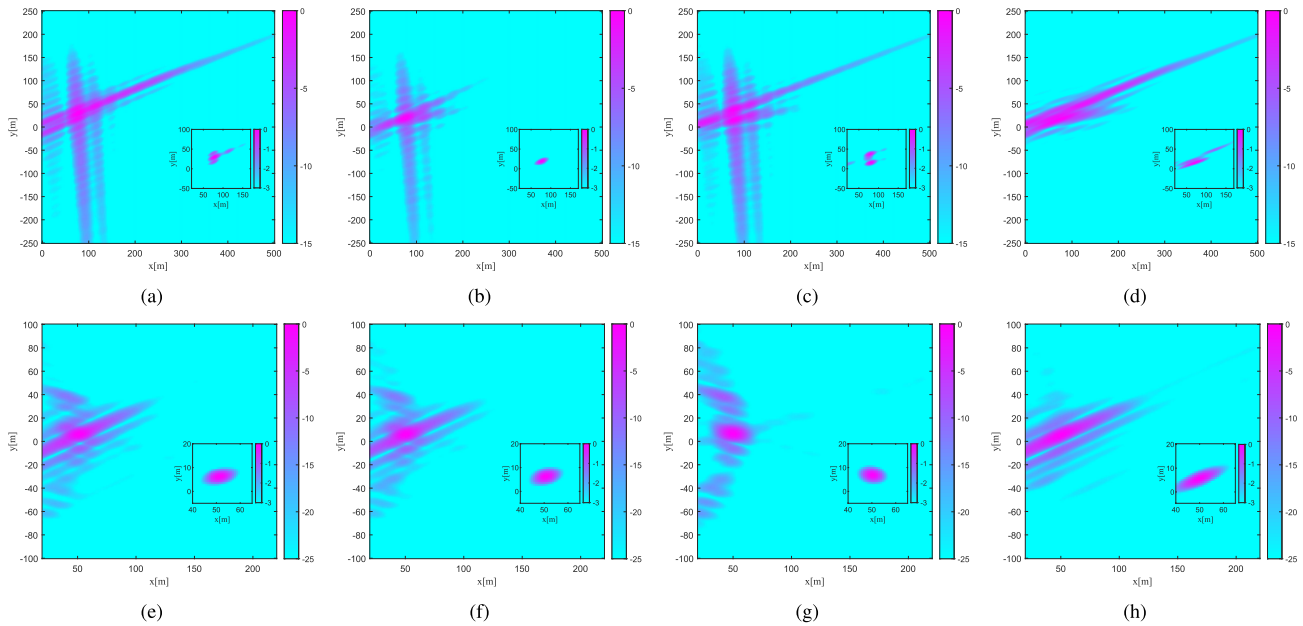


Fig. 8. Image fusion from a combination of three satellites. (a) G15–G18–G24. (b) G15–G18–G32. (c) G15–G24–G32. (d) G18–G24–G32. (e) G17–G23–G01. (f) G17–G23–G03. (g) G17–G01–G03. (h) G23–G01–G03.

TABLE II
EXPERIMENTAL RESULTS OF GPS-TAG

Satellite Signal	Satellite combination pair	Theoretical Angle difference	Actual Angle difference	Resolution unit area
GPS L1 C/A code	G15-G18	74.5°	75°	$1074m^2$
	G15-G24	84.5°	79°	$1021m^2$
	G15-G32	85.9°	87°	$899m^2$
	G18-G24	10.0°	8°	$5926m^2$
	G18-G32	19.6°	14°	$4939m^2$
	G24-G32	9.6°	9°	$7425m^2$
GPS L5 C/A code	G17-G23	53.3°	53°	$201m^2$
	G17-G01	37.0°	38°	$154m^2$
	G17-G03	56.9°	65°	$132m^2$
	G23-G01	16.3°	15°	$1017m^2$
	G23-G03	3.6°	8°	$452m^2$
	G01-G03	19.9°	27°	$380m^2$

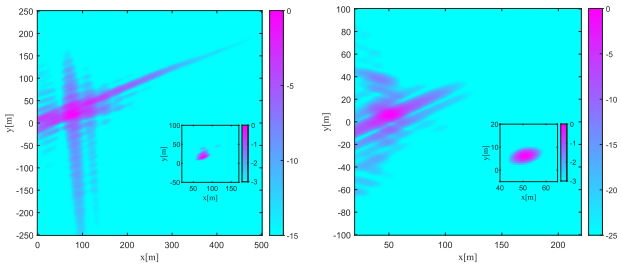


Fig. 9. Image fusion from a combination of four satellites. (a) GPS L1 (G15–G18–G24–G32). (b) GPS L5 (G17–G23–G01–G03).

Fig. 10 presents the relationship between computational load and the area of the minimum resolution unit corresponding to various satellite combinations. Fig. 10(a) shows the results for the GPS L1 C/A code signals, while Fig. 10(b) presents the results for the GPS L5 C/A code signals. Within the GNSS-R SAR imaging framework, imaging processing necessitates the highest computation effort. Assuming that the computational load for a single BP imaging operation is denoted as N, the bar chart in the figure depicts the computation amounts associated with different combinations of satellites. Additionally, the line plots demonstrate the area of the minimum resolution unit corresponding to these combinations. Notably, the area of the minimum resolution unit does not decrease as the number of satellites increases, attributed to the presence of ghost targets during the multisatellite fusion process. Among the evaluated combinations, the fusion results of satellites G15 and G32 in Fig. 10(a) and the fusion results of satellites G17 and G03 in Fig. 10(b) have the smallest resolution per unit area and the lowest computational burden. Additionally, the imaging results of these two pairs are the closest to circular, resembling the theoretical imaging results of point targets the most. This also indicates that when the angle between the optimal resolution directions of the two satellites approaches 90°, the fusion imaging result becomes closest to the true image of the target. Consequently, selecting two optimally matched satellites for SAR image fusion processing

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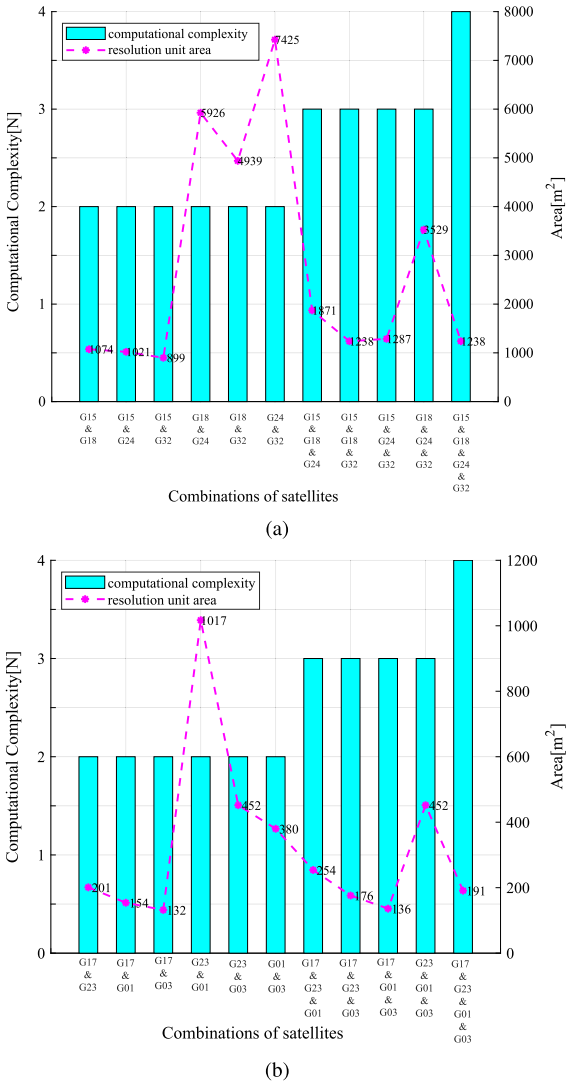


Fig. 10. Results of computational complexity and minimum resolution unit area corresponding to different satellite combinations. (a) GPS L1. (b) GPS L5.

not only minimizes computational demands, but also enhances resolution effectiveness.

The experimental results obtained from GPS-TAG indicate that the proposed method is effective for processing real GPS L1 C/A code and GPS L5 C/A code signals, and the selection principle for optimal matching satellite pairs has been reasonably validated.

C. Scene Imaging Experiment and Analysis

This section examines the effect of the proposed method on imaging actual ground scenes. The equipment used in the experiment is shown in Fig. 11(a). Since the polarization direction of GNSS signals transmitted by satellites changes upon scene reflection, the reflected signals exhibit various polarization types, primarily consisting of left-handed circular polarization. To effectively receive the weak reflected signals, a left-handed circularly polarized high-gain directional antenna is employed as the antenna for the GNSS receiver's reflection channel [36], coupled with a low-noise amplifier (LNA). The direct wave channel of GNSS receiver still uses a GPS

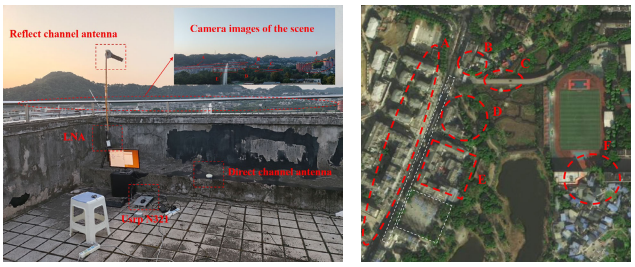


Fig. 11. Photographs of the environment and equipment used in the scene imaging experiment. (a) Experimental setup and camera images of the scene. (b) Satellite photographs of the detection area.

omnidirectional antenna. The receiver is configured with a sampling rate of 49.152 MHz, an acquisition time of 120 s, and a data duration for SAR imaging set to 80 s. The experimental location mirrors that of the GPS-TAG experiment; however, the area detected by the antenna is situated on the North side of the Yifu Building. With the receiver designated as the origin, the distance range of the imaging area extends from 300 to 850 m, while the azimuth range covers −300 to 250 m. The on-site photographs of the detection area are depicted in Fig. 11(a), with the main buildings in the scene indicated by red curves and letters. Area A comprises a contiguous building with a regular arrangement, while areas B, D, and F consist of several isolated buildings. Area C features a viaduct, and area E is characterized by a building with a square outline. Additionally, Fig. 11(b) displays the satellite image of the area, where the building region outlined by the red curve corresponds to Fig. 11(a), and the rectangular and square areas enclosed by the white dotted lines represent the streets and squares within the scene, respectively.

The GNSS software receiver is utilized to capture and track the original data, successfully tracking a total of three GPS satellites: G03, G28, and G11. Following the conversion of coordinates, the positional and velocity information of each satellite is substituted into (28) to perform angle searches. The optimal resolution angles obtained for satellites G03, G28, and G11 are found to be 157°, 78°, and 41°, respectively. According to the selection principle, the angle between the optimal resolution angles of G03 and G28 is closest to 90°, establishing G03 and G28 as the optimal matching satellite pair. SAR imaging is subsequently conducted on G03 and G28, following the methodologies presented in the literature [10], with the resulting images displayed in Fig. 12(a) and (b). Due to the proximity of imaging regions A–E, coupled with the limited bandwidth of the GPS L1 C/A code signals, the resolution remains insufficient. Consequently, the imaging results for each building in Fig. 12(a) and (b) are aliased together, rendering them indistinguishable and making it impossible to correlate with the actual scene. In the field of SAR imaging, as referenced in [37], [38], and [39], aligning SAR images with optical images or satellite imagery of the actual scene constitutes an intuitive method for evaluating the imaging effect. Fig. 12(d) and (e) shows the results of matching SAR images with satellite map photographs, and the satellite images of the scenes in this experiment are obtained from Baidu maps.

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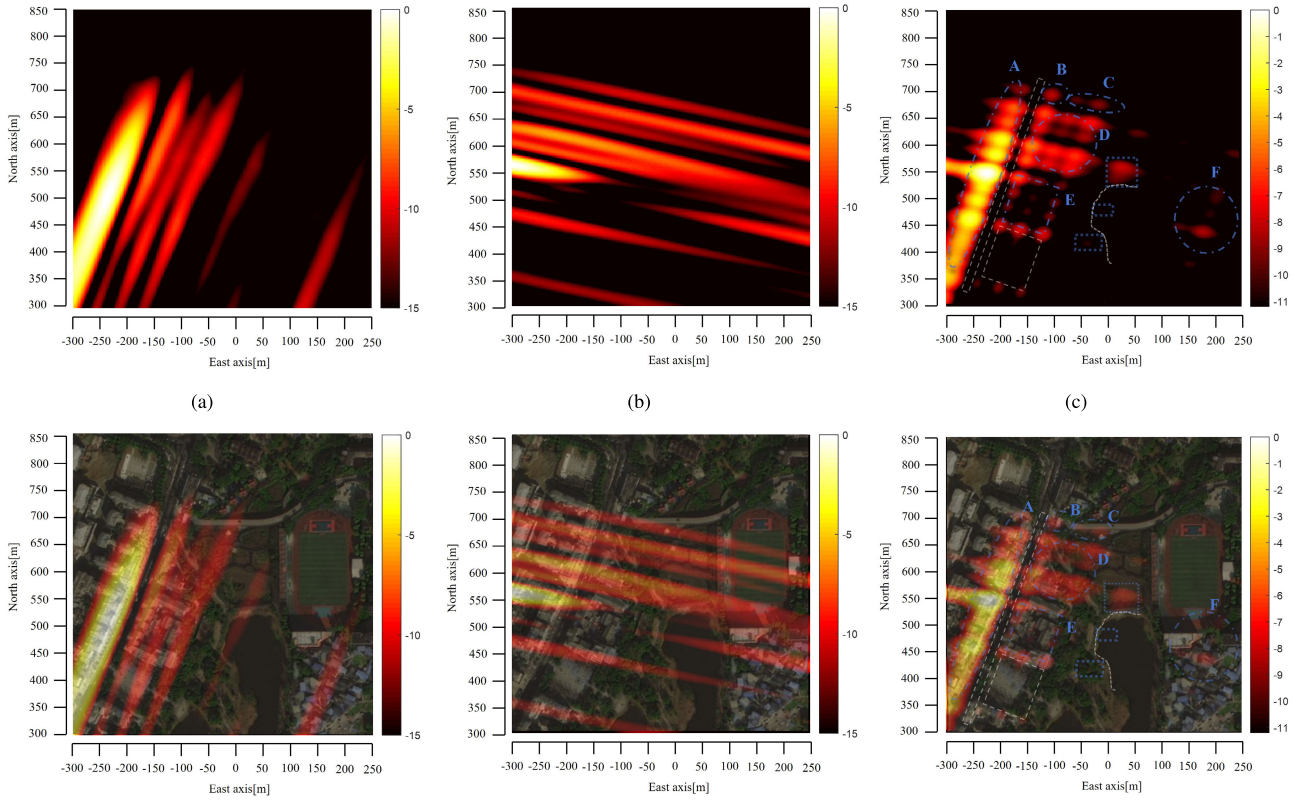


Fig. 12. Results of SAR imaging and its map photograph matching. (a) BP imaging result for G03. (b) BP imaging result for G28. (c) Fusion results for BP imaging of G03 and G28. (d) Matching of G03 SAR images with map photographs. (e) Matching of G28 SAR images with map photographs. (f) Matching the fused SAR images with map photographs.

The result of the fusion processing of Fig. 12(a) and (b) is presented in Fig. 12(c). Within this figure, the principal structures are demarcated by blue dashed lines, and the rectangular road and square plaza, indicated by white dashed lines, are also clearly visible. Furthermore, the three isolated imaging points along the white curve align with the outline of the lake shore as it appears in the actual scene. Fig. 12(f) demonstrates the matching results between the fusion imaging and satellite photographs. It is observable from the figure that structures, which were previously indistinct, become clearly discernible following the image fusion processing, signifying an improvement in imaging resolution.

The SAR imaging experiment of ground scenes employing the GPS L1 C/A code signals indicates that, under identical conditions, the SAR imaging resolution of a single satellite is inadequate, and the resultant images do not correspond with the actual scene. The method proposed in this article enhances the image resolution characteristics of the entire system. After image fusion processing, the new imaging results are found to be in close agreement with the actual scene.

VI. Conclusion

This article presents a GPS SAR imaging fusion method that utilizes the CRLB to identify two optimally matched satellites. By establishing the relationship between the GNSS signals ambiguity function and Fisher information, the coordinates of ground targets are defined in polar form, and the CRLB for parameter estimation of ground targets is derived. The CRLB expression facilitates the calculation of the optimal resolution direction for ground target detection based on satellite orbital parameters, thereby providing a theoretical foundation for the selection of satellites in GNSS-R SAR imaging fusion. The proposed method enhances the resolution of the GNSS-R SAR system while enabling the preselection of the best-matching satellite pairs prior to dual-satellite imaging fusion. This approach effectively simplifies the process, thereby avoiding redundant operations and improving the overall efficiency of SAR imaging. The simulation and real experimental results demonstrate the effectiveness of the proposed method.

Furthermore, it is important to state that the technology developed in this study for the GNSS-R SAR system with ground-fixed receiver is not restricted to GPS L1 C/A code signals. Rather, it has the potential for broader applicability to all GNSS signals, especially those with small bandwidths.

Appendix
Proofs of Decoupled Ambiguity Function

To decouple f from η in (10), an approximation of $\overrightarrow{\beta}(\eta)$ is considered, along with a Taylor expansion of $\overrightarrow{\beta}(\eta)$ at the midpoint η_0 in slow time. $\overrightarrow{\beta}(\eta)$ can be estimated as follows:

$$\overrightarrow{\beta}(\eta) \simeq \overrightarrow{\beta}(\eta_0) + (\eta - \eta_0)\frac{d\overrightarrow{\beta}}{d\eta}(\eta_0). \tag{A.1}$$

In (A.1), decoupling can be achieved if $\overrightarrow{\beta}(\eta)$ in the first exponential integral term on the right side of the equation

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is replaced by $\overrightarrow{\beta}(\eta_0)$. It is obvious that the approximation of $\overrightarrow{\beta}(\eta)$ will bring the largest deviation $\Delta_{\max}$ when $\eta = T$. If the effect of $\Delta_{\max}$ is negligible, it is equivalent to the following equation:

$$\int \exp\left(-j\frac{2\pi f}{c}\left(\overrightarrow{OP}\cdot\left(\overrightarrow{\beta}(\eta_0)+\eta\frac{d\overrightarrow{\beta}}{d\eta}(\eta_0)\right)\right)\right)df - \int \exp\left(-j\frac{2\pi f}{c}\left(\overrightarrow{OP}\cdot\left(\overrightarrow{\beta}(\eta_0)+T\frac{d\overrightarrow{\beta}}{d\eta}(\eta_0)\right)\right)\right)df = 0. \tag{A.2}$$

Obviously, the tenability of (A.2) is equivalent to the tenability of the following equation, which can be denoted as:

$$\int \exp\left(-j\frac{2\pi f}{c}\overrightarrow{OP}\cdot\left(T\frac{d\overrightarrow{\beta}}{d\eta}(\eta_0)\right)\right)df = 1. \tag{A.3}$$

Assuming that B_{gnss} is the bandwidth, $f \in [0, B_{\text{gnss}}]$, the solution of (A.3) is given by

$$B_{\text{gnss}} = \frac{c}{2T\overrightarrow{OP}\cdot\frac{d\overrightarrow{\beta}}{d\eta}(\eta_0)}. \tag{A.4}$$

Therefore, f and η can be decoupled if the system parameters satisfy the condition $c/(2T\overrightarrow{OP}\cdot(d\overrightarrow{\beta}/d\eta)(\eta_0)) \geq B_{\text{gnss}}$. Under this condition, (10) can be rewritten as

$$\chi(\overrightarrow{OP}) = K\int\int |\widetilde{P}(f-f_c)|^2 \exp\left(-j\frac{2\pi f}{c}\overrightarrow{OP}\cdot\overrightarrow{\beta}(\eta)\right) \times \exp\left(j\frac{2\pi f_c}{c}\overrightarrow{OP}\cdot\overrightarrow{\beta}(\eta)\right)df\,d\eta. \tag{A.5}$$

When the distance from the satellite to the receiver and the target is much greater than the distance from the receiver to the target, τ_d and f_d can be expressed as [22]

$$\tau_d = \frac{\overrightarrow{OP}\cdot\overrightarrow{\beta}(\eta_0)}{c} \tag{A.6}$$

$$f_d = f_0 \times \frac{\overrightarrow{OP}\cdot\frac{d\overrightarrow{\beta}(\eta_0)}{d\eta}}{c}. \tag{A.7}$$

Using the properties of the Fourier transform [40], [41], (A.5) represented in the time domain can be expressed in the frequency domain as

$$\chi(\overrightarrow{OP}) = K\times B\frac{\sin(\pi B\tau_d)}{\pi B\tau_d}\times T\frac{\sin(\pi T f_d)}{\pi T f_d}. \tag{A.8}$$

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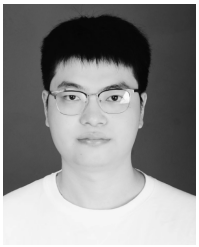
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